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Multivariate Classification and Differential Expression Analysis of Whole Blood RNA-seq

Data in Type 1 Diabetes

1. Introduction

Type 1 diabetes (T1D) is an autoimmune disease that results from beta cell destruction in the pancreas, resulting in insulin deficiency.¹ The biological question I am hoping to answer is: To what extent can whole blood gene expression profiles distinguish individuals with T1D from healthy controls, and which specific genetic markers drive this differentiation across univariate and multivariate statistical models? To answer this question, my analysis goals after data preparation are to first perform a PCA analysis to see whether the samples cluster by disease status in the principal component space. This step would be mainly for visualization purposes. Next, I would like to perform gene-by-gene inference to test the association of gene expression with disease status. This will be completed using a standard t-test, comparing the mean expression of the healthy group and the mean expression of the affected individuals for each gene. Finally, I want to see how well I can classify samples using a random forest with leave-one-out cross validation (LOOCV). The results of this random forest analysis will illuminate how well gene

expression as a multivariate pattern can classify healthy vs. affected individuals within the dataset.

2. Methods

2.1. Data Availability

The dataset comes from a 2020 study by Valentim et al.¹ investigating transcriptomic signatures in type 1 diabetes (T1D). It contains 82 whole blood RNA-seq samples, where 39 samples are from patients with T1D and the remaining 43 samples are from healthy donors. The samples were collected in Paris, France and the RNA-seq libraries were prepared using the TruSeq Stranded mRNA preparation kit (Illumina). Sequencing was performed while adhering to a pair-end 2x75 bp protocol on NextSeq 500 or HiSeq 4000 (Illumina). The resulting raw counts dataset includes expression levels for 39,376 gene IDs.

2.2. Data Preparation

Before performing any statistical analyses, the dataset was read into R as a count matrix and a metadata object that defined healthy samples and T1D samples was created. Next, genes with low counts were filtered out to remove the noise of any genes with low or inconsistent expression. The filtering rule was to only keep genes with at least 10 counts in a minimum of 39 samples (corresponding to the size of the smallest experimental group, T1D). After doing so, there were a total of 18,015 genes remaining. Next, a `DESeq` object was created using the metadata and the filtered counts. Finally, Variance Stabilizing Transformation (VST) was performed to decouple the variance from the mean, handling both normalization and log-transformation of the remaining genes.

2.3. Principal Component Analysis (PCA)

To perform the principal component analysis, the `scikit-learn` library in Python was utilized on the VST-transformed data containing all 18,015 genes remaining from the filtering step. First, the data was transposed so samples became rows and genes became columns. Next, the data was standardized using `StandardScaler()`. Then, the `PCA()` function was applied and the results were visualized in a scatterplot. Next, the silhouette method was used to determine the optimal number of clusters for the data, and a second PCA scatterplot was made to visualize the data with clusters delineated on PCA space.

2.4. Gene-by-Gene Inference (Standard t-test)

To identify individual genes associated with T1D status, a two-sample t-test was performed for each of the 18,015 genes using the built-in R function `t.test()`. To account for the high rate of false positives inherent in testing thousands of genes simultaneously, p-values were adjusted using the Benjamini-Hochberg (BH) procedure using the `p.adjust()` function in R. A gene was considered statistically significant if its adjusted p-value was less than 0.05.

2.5. Random Forest (with LOOCV)

To construct a classification model using random forest with leave-one-out-cross-validation (LOOCV), the `scikit-learn` library in Python was utilized on the VST-transformed data containing all 18,015 genes remaining from the filtering step. The classifier was configured using 500 decision trees and the number of features to be considered at each split was set to equal the square root of the number of genes (~ 134). In this LOOCV approach, the model is trained 82 times, each time leaving out one sample as the test set while using the remaining 81 samples for training, ensuring that every sample is evaluated exactly once.

3. Results

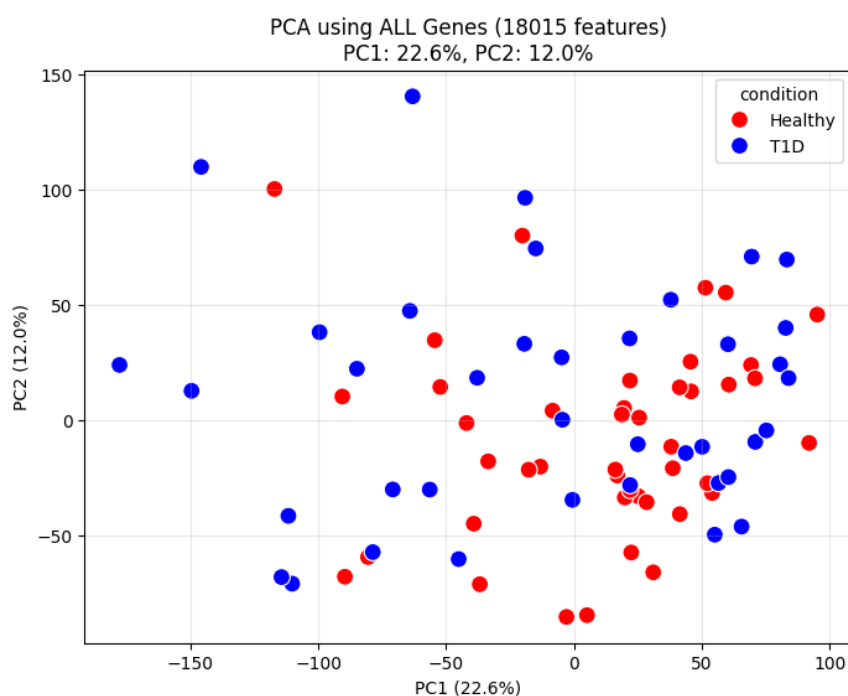


Figure 1: Resulting principal component analysis (PCA) plot from using the PCA function in the `scikit-learn` library on the VST-transformed data.

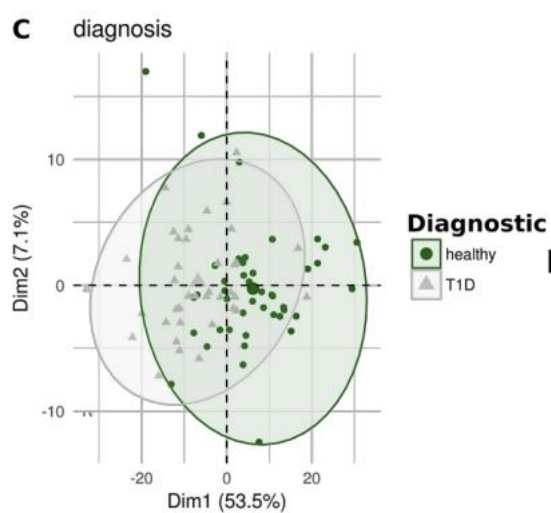


Figure 2: PCA plot from Valentim et al. paper¹.

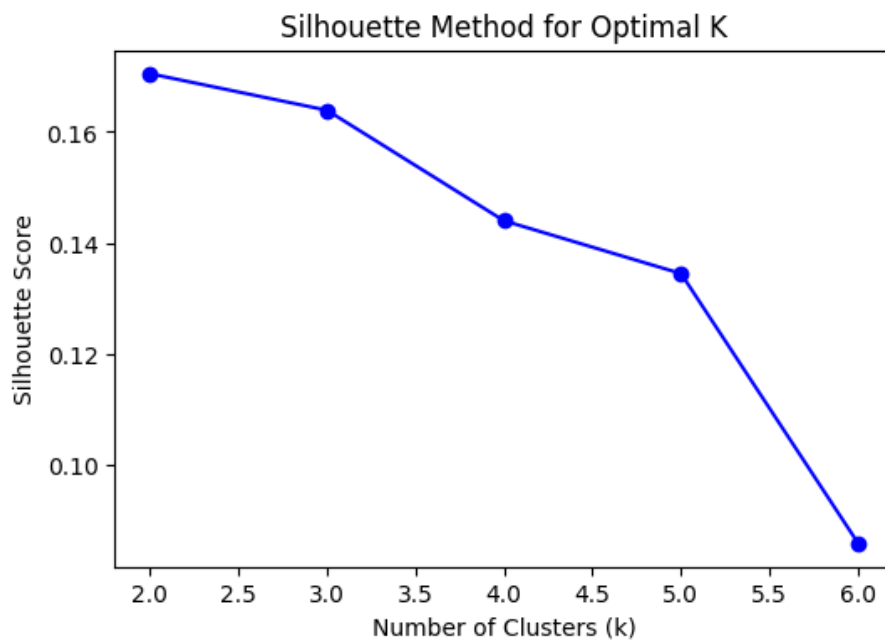


Figure 3: Plot showing optimal k value for PCA clustering analysis.

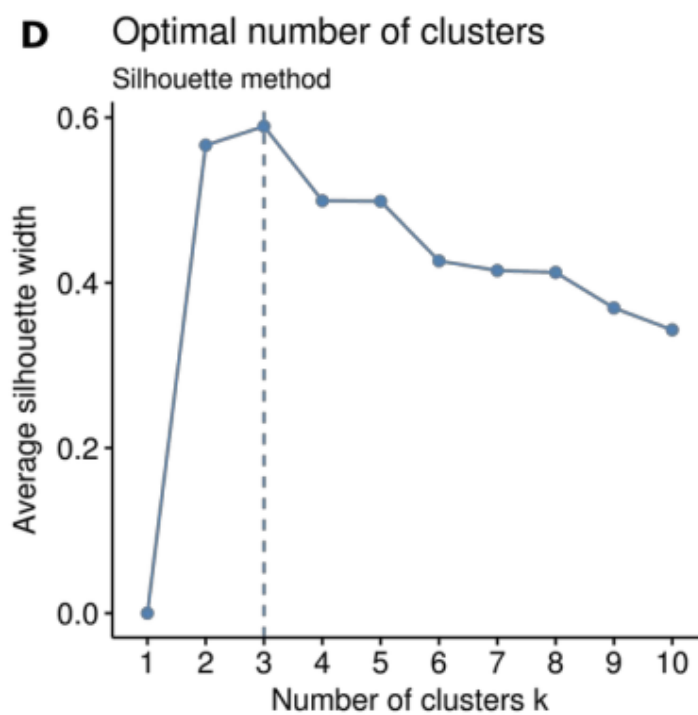


Figure 4: Optimal k plot from Valentim et al. paper¹.

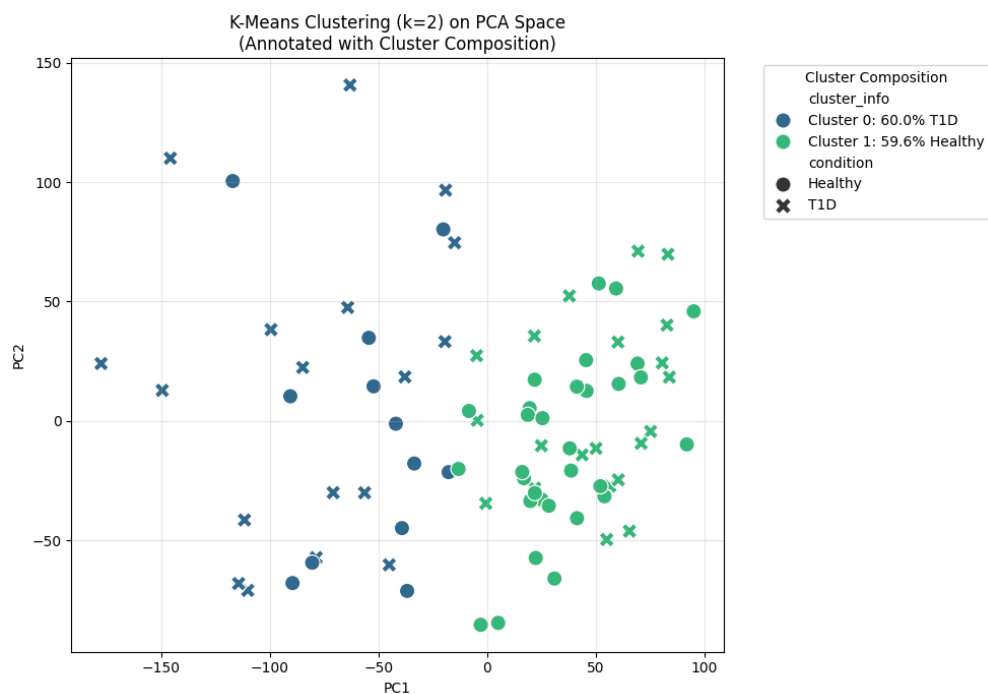


Figure 5: Scatterplot visualizing the data with k=2 clusters delineated on PCA space.

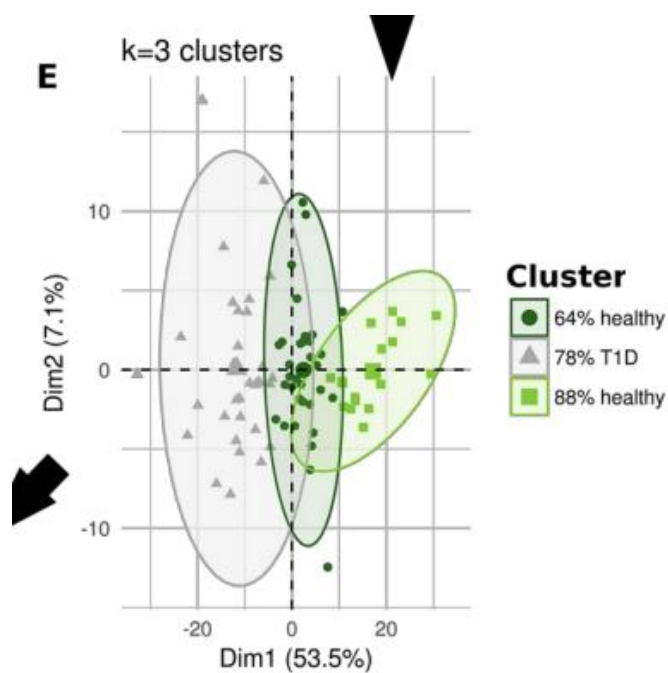


Figure 6: Scatterplot with k=3 clusters delineated in PCA space from Valentim et al. paper¹.

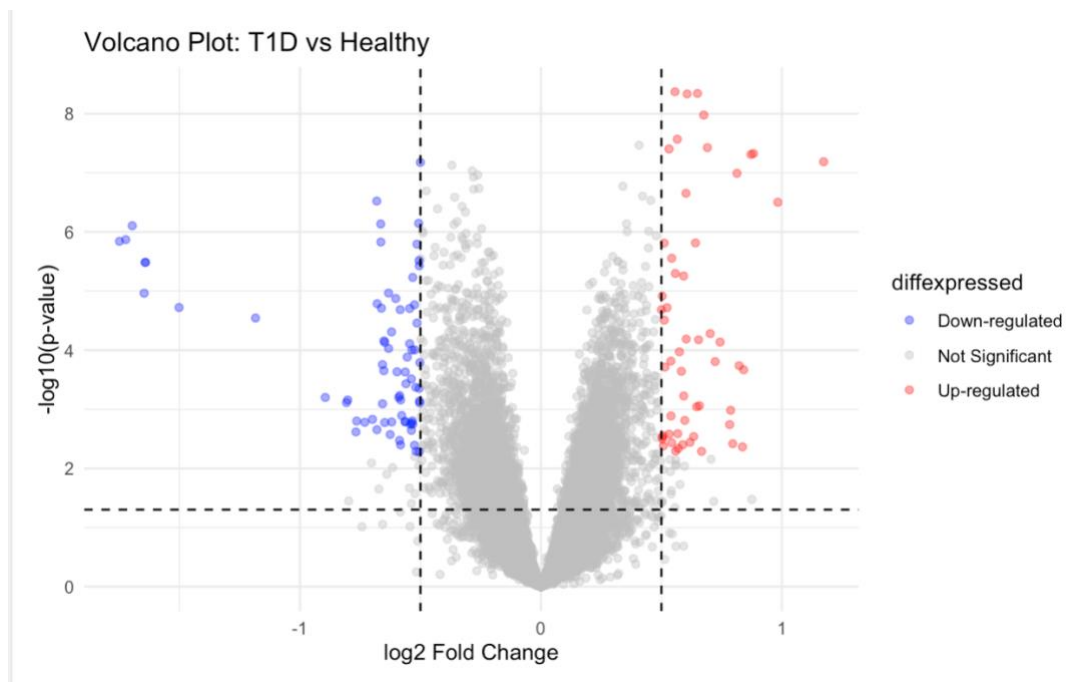


Figure 7: Differential Gene Expression in T1D vs. Healthy Controls.

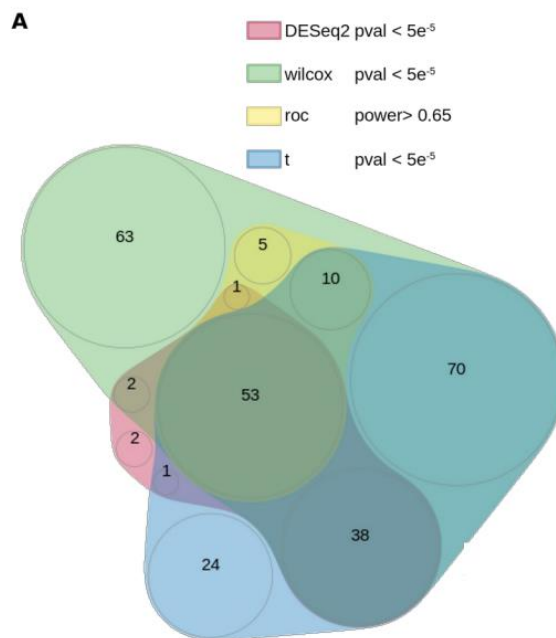


Figure 8: Differential analysis performed over 4 different tests from Valentim et al. paper¹.

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--- Results for ALL 18,015 Filtered Genes ---
Accuracy: 0.8049
Kappa: 0.6098

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Figure 9: Results of classification model using random forest with LOOCV via `scikit-learn` library.

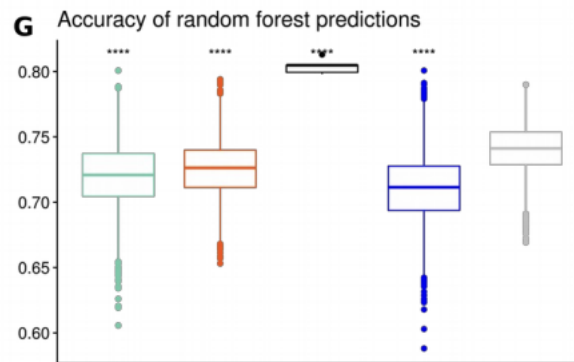


Figure 10: Random forest prediction results from Valentim et al. paper¹.

4. Discussion

Figure 1 shows the results of my Principal Component Analysis. The blue dots represent individuals with type 1 diabetes, and the red dots represent healthy individuals. Similar to the results from the Valentim et al. paper¹ shown in figure 2, there is no clear separation between healthy individuals and those with T1D. However, it appears that in both figures, points to the far left belong to individuals with T1D. Figures 3 and 4 show the optimal number of clusters for the data from my analysis and the Valentim et al. paper¹, respectively. My optimal k value was 2, while the authors of the paper had k=3. This discrepancy could be because I filtered the data to exclude genes with severely low counts while the Valentim et al. paper¹ authors used the full dataset. Figure 5 visualizes the data with k=2 clusters in PCA space, and figure 6 from the Valentim et al. paper¹ delineates the data in PCA space with k=3 clusters. Both figures have clusters where

there is no tie between the number of samples within each cluster, however figure 6 has more healthy individuals within its clusters. These results indicate that the heterogeneity of T1D makes it difficult to differentiate signals from patients with the condition versus those without it.

Figure 7 is a volcano plot showing the transcriptomic differences between individuals with T1D and the healthy controls. The horizontal dotted line represents the p-value threshold (0.05), the vertical dotted lines indicate the fold-change walls which represent the biological impact threshold, the grey dots are insignificant genes (17,888 total), the red dots are significant genes that are up-regulated (55 total), and the blue dots represent significant genes that are down-regulated (72 total). This sums up to a total of 127 significant genes out of 18,015 total genes after filtering. We compare this to figure 8, where the Valentim et al. paper¹ authors found a total of 196 significant genes at a $5e-5$ significance threshold. Despite the strict statistical thresholds applied to the unfiltered data by Valentim et al.¹, my analysis of the filtered dataset yielded highly concordant results. This suggests that the core transcriptomic signal of T1D remains robust across different preprocessing pipelines and confirms the validity of my filtered feature set.

Figure 9 shows the results of the random forest with LOOCV classification model. The model achieved an accuracy of $\sim 80.5\%$, which serves as an indication that the multivariate gene expression pattern is a reliable predictor of T1D status. This result is similar to the $\sim 80\%$ accuracy achieved by the random forest model used by the Valentim

et al. paper¹ authors shown in Figure 10. Figure 9 also shows a (rounded) Cohen's Kappa value of 0.61, which signifies that the accuracy of my model is not due to random chance.

5. Conclusions

In conclusion, this study successfully replicated and validated the key findings of Valentim et al.¹, demonstrating that while the whole-blood transcriptomic signature of Type 1 Diabetes is characterized by significant heterogeneity, it contains a robust and reproducible signal. The PCA and clustering results highlight the subtle nature of the disease's impact on the genome, where the lack of clear group separation suggests that T1D patients exist on a transcriptomic spectrum rather than in a binary state that can be easily distinguished from healthy individuals. However, the identification of 127 high-confidence differentially expressed genes, which proved to be consistent with the original study's findings despite varying statistical thresholds, demonstrates that a "core" signature of T1D remains detectable. Finally, the ~80.5% accuracy and substantial Cohen's Kappa (~0.61) achieved by the random forest model confirm that these gene expression patterns are not only statistically significant but also possess meaningful predictive value.

Collectively, these results suggest that filtered, high-quality transcriptomic data can serve as a reliable foundation for understanding the molecular drivers of T1D and developing future diagnostic biomarkers.

6. Acknowledgements

AI assistance (Google Gemini) was used to help debug and refine portions of the R code used for data processing.

References

1. Valentim, F. L., Konza, O., Velasquez, A., Saadawi, A., Patel, N., Lorenzon, R., Tchitchek, N., Mariotti-Ferrandiz, E., Klatzmann, D., & Six, A. (2025). *Transcriptomic module fingerprint reveals heterogeneity of whole blood transcriptome in type 1 diabetic patients*. Bioinformatics. <https://doi.org/10.1101/2025.11.19.689306>